

TTL - transistor - transistor Logic
CMOS - complementary Metal - Oxide semiconductors

Boolean algebra order priority:

()
AND
XOR
OR

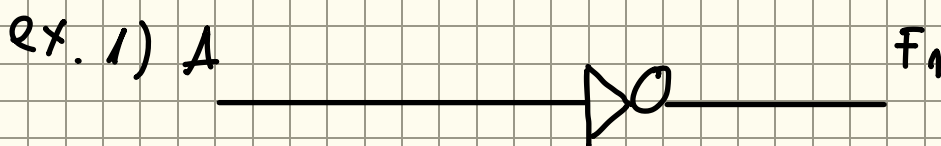
High
↓
Low

AND = $\cdot$
XOR = $\oplus$
OR = $+$

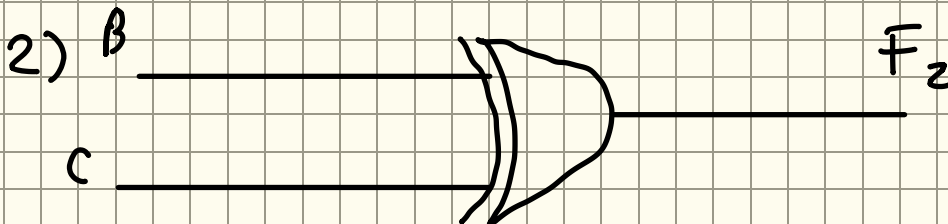
De Morgan Law:

$$\overline{a \cdot b} = \bar{a} + \bar{b}$$

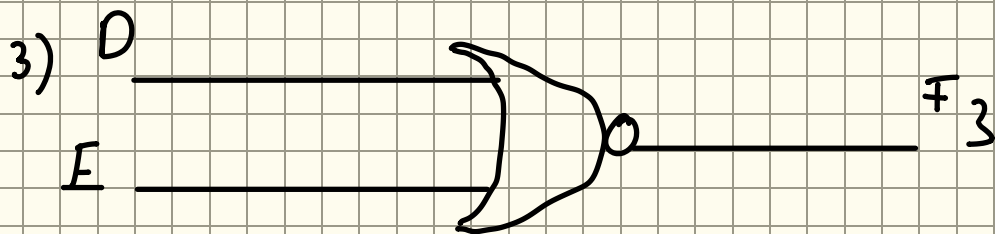
$$\overline{a + b} = \bar{a} \cdot \bar{b}$$



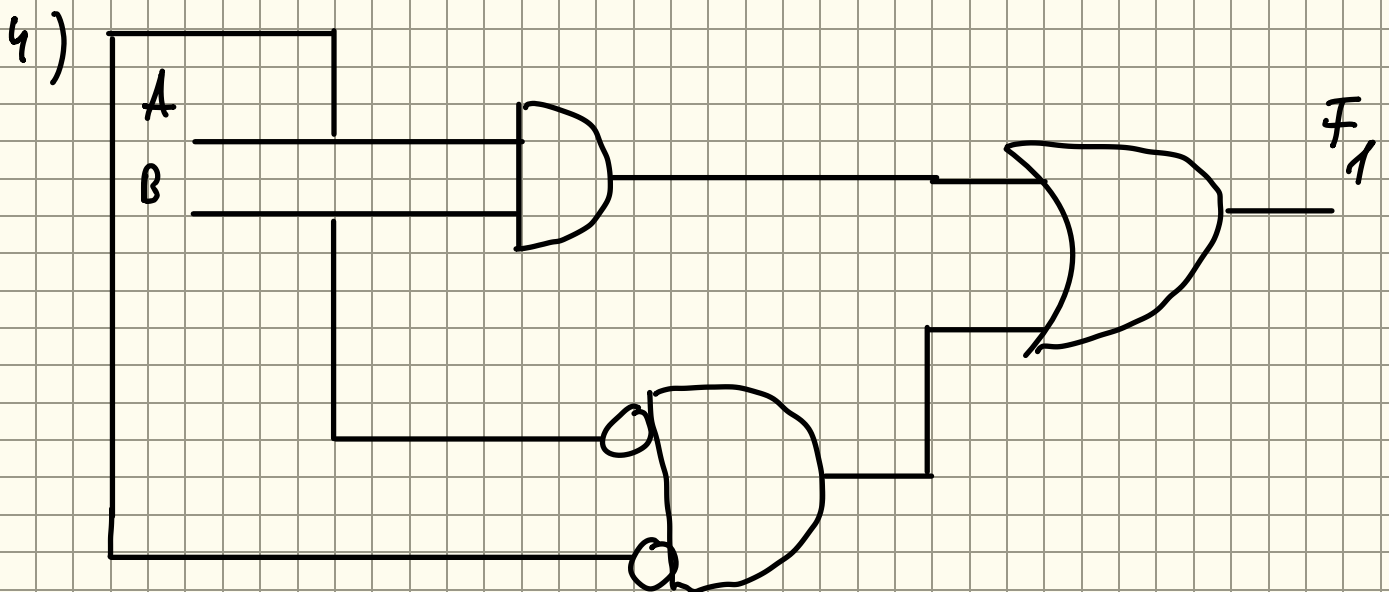
A	F ₁
0	1
1	0



B	C	F ₂
0	0	0
0	1	1
1	0	1
1	1	0



D	E	F ₃
0	0	1
0	1	0
1	0	0
1	1	0



A	B	F ₁
0	0	1
1	0	0
0	1	0
1	1	1

$F_1 = \text{XNOR}$

5) $\bar{a} \cdot b + \bar{c} \Rightarrow$

$\bar{a} =$ $\begin{matrix} 1 \\ 1 \\ 1 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{matrix}$ $\bar{c} =$ $\begin{matrix} 1 \\ 0 \\ 1 \\ 0 \\ 1 \\ 0 \\ 1 \\ 0 \end{matrix}$

a	b	c	F ₁
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	
1	0	1	0
1	1	0	1

1111

$$6) a \cdot c + \overline{b \cdot c}$$

a